

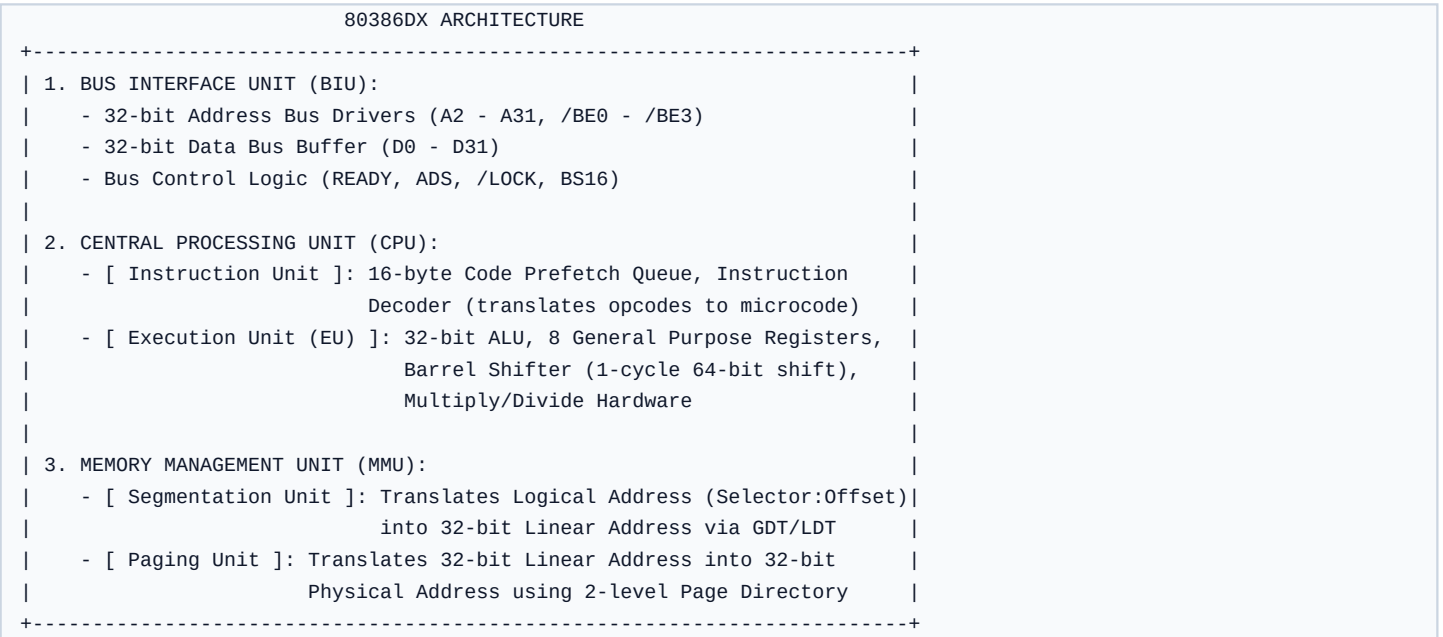
80386DX Architecture & Functional Units -- MPCA Module 5

Definition: The Intel 80386DX is a true 32-bit microprocessor with a 32-bit internal architecture, a 32-bit Data Bus, and a 32-bit Address Bus, capable of addressing 4 GB of physical memory and up to 64 Terabytes (TB) of virtual memory.

1. Detailed Technical Explanation

1. Functional Block Diagram of 80386DX

The 80386DX architecture is organized into three major sections containing six distinct functional units:



2. Address Spaces in 80386DX

1. Physical Address Space: $2^{32} = 4$ GB of physical RAM directly addressable.

2. Vir

3. Core Concepts & Memory Keywords

- * 32-bit Data & Address Bus: Transfers 32-bit dwords in a single 2-clock bus cycle.

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4. Must-Write Points for Exams

- * The 80386DX has a 32-bit data bus and 32-bit address bus (unlike 80386SX which has a 16-bit data bus and 24-bit address bus).

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5. Quick Recall Flow

80386DX (32-bit) -> 3 Sections (BIU, CPU, MMU) -> Segmentation (Logical to Linear) -> Paging (Linear to Physical) -> 4GB RAM / 64TB Virtual